

Optimization of the thermoelectric figure of merit in the conducting polymer poly(3,4-ethylenedioxythiophene)

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Thermoelectric generators (TEGs) transform a heat flow into electricity. Thermoelectric materials are being investigated for electricity production from waste heat (co-generation) and natural heat sources. For temperatures below 200 °C, the best commercially available inorganic semiconductors are bismuth telluride (Bi₂Te₃)-based alloys, which possess a figure of merit ZT close to one¹. Most of the recently discovered thermoelectric materials with ZT > 2 exhibit one common property, namely their low lattice thermal conductivities^{2,3}. Nevertheless, a high ZT value is not enough to create a viable technology platform for energy harvesting. To generate electricity from large volumes of warm fluids, heat exchangers must be functionalized with TEGs. This requires thermoelectric materials that are readily synthesized, air stable, environmentally friendly and solution processable to create patterns on large areas. Here we show that conducting polymers might be capable of meeting these demands. The accurate control of the oxidation level in poly(3,4-ethylenedioxythiophene) (PEDOT) combined with its low intrinsic thermal conductivity ($\lambda = 0.37 \text{ W m}^{-1} \text{ K}^{-1}$) yields a ZT = 0.25 at room temperature that approaches the values required for efficient devices.

Unlike their inorganic counterparts, organic semiconductors have not been thoroughly investigated in terms of their potential for thermoelectric applications, primarily because of their relatively low power factor $\alpha^2\sigma$ (numerator of ZT). Interestingly, conducting polymers possess a low intrinsic thermal conductivity, which varies between $0.2 \text{ W m}^{-1} \text{ K}^{-1}$ for poly(3,4-ethylenedioxythiophene) complexed with polystyrene sulphonic acid (PEDOT:PSS) and $0.7 \text{ W m}^{-1} \text{ K}^{-1}$ for highly doped polyacetylene^{4–6}. These low thermal conductivities are comparable to inorganic and organic amorphous materials⁷. For this class of material, most of the phonons have mean free paths reduced to the minimum separation between equivalent structural units, for instance the interchain distance in polymers defined by weak van der Waals interactions. Organic materials provide other noteworthy advantages in the form of mechanical flexibility, low-cost synthesis, and solution processability over large areas. The applications foreseen for large-area flexible thermogenerators and Peltier coolers are most probably inconceivable with inorganic materials, as they are both brittle and expensive.

Recent studies show that conducting polymers including polyaniline, polypyrrole and poly(3,4-ethylenedioxythiophene) (PEDOT) exhibit attractive thermoelectric properties^{8,9}, with

a record power factor of about $19 \mu\text{W m}^{-1} \text{ K}^{-2}$ reported for polycarbazole derivatives¹⁰. PEDOT is one of the most extensively explored conducting polymers for use in thin-film conductive coatings and as the electrode material in organic electronics¹¹. It is well known for its environmental stability, wide range of conductivity and solution processability. The electrical conductivity of conjugated polymers depends significantly on the morphology, microstructure and the number of counterions that balance the positive doping charge carried by the conjugated polymer chains¹². A non-optimized ZT value at room temperature for PEDOT:PSS of 1.75×10^{-3} has been reported⁵, which is approximately a factor of a thousand lower than that for Bi₂Te₃ alloys. The polyanion (PSS) can be replaced by small anions, such as tosylate (Tos), thus preventing the excess of insulating phase generated with polyanions¹², and as a result PEDOT–Tos films can reach electrical conductivity values exceeding $1,000 \text{ S cm}^{-1}$ (ref. 13).

In this work, we optimize the power factor $\alpha^2\sigma$ of the conducting polymer PEDOT–Tos by controlling its oxidation level. With the low intrinsic thermal conductivity of conducting polymers, ZT reaches 0.25 at room temperature. The relatively high figure of merit of this air-stable conducting polymer provides promise for large-area flexible thermogenerators.

PEDOT–Tos is polymerized directly by mixing the EDOT monomers and an oxidative solution of iron(III) *tris*-p-toluenesulphonate. The electrical conductivity of freshly polymerized PEDOT–Tos thin films measured by the standard four-point probe method is around 300 S cm^{-1} . The presence of crystalline nano-domains characterized by π – π interactions between adjacent polymer chains (interchain distance about $3.4 \text{ \AA}$) with the Tos counterions outside the chain stacks explains the high conductivity of PEDOT–Tos (ref. 14). The Seebeck coefficient of the polymer is measured to be about $40 \mu\text{V K}^{-1}$. Consequently, the power factor of the thin films ($S = \alpha^2\sigma$) is $38 \mu\text{W m}^{-1} \text{ K}^{-2}$ at room temperature. Since the electrical conductivity decreases at lower oxidation levels, whereas the Seebeck coefficient tends to increase, the power factor of organic semiconductors reaches its optimum at a specific oxidation state. To track this optimum and to enhance the power factor value, the polymer is reduced in an inert atmosphere with tetrakis(dimethylamino)ethylene (TDAE) to alter the polymer's electronic structure. As soon as the polymer is exposed to a vapour of TDAE molecules, electron transfer takes place from the reducing agent to the PEDOT–Tos, thus resulting in the transformation of positively charged (oxidized) polymer chains into neutral ones¹⁵. After the electron transfer, TDAE²⁺

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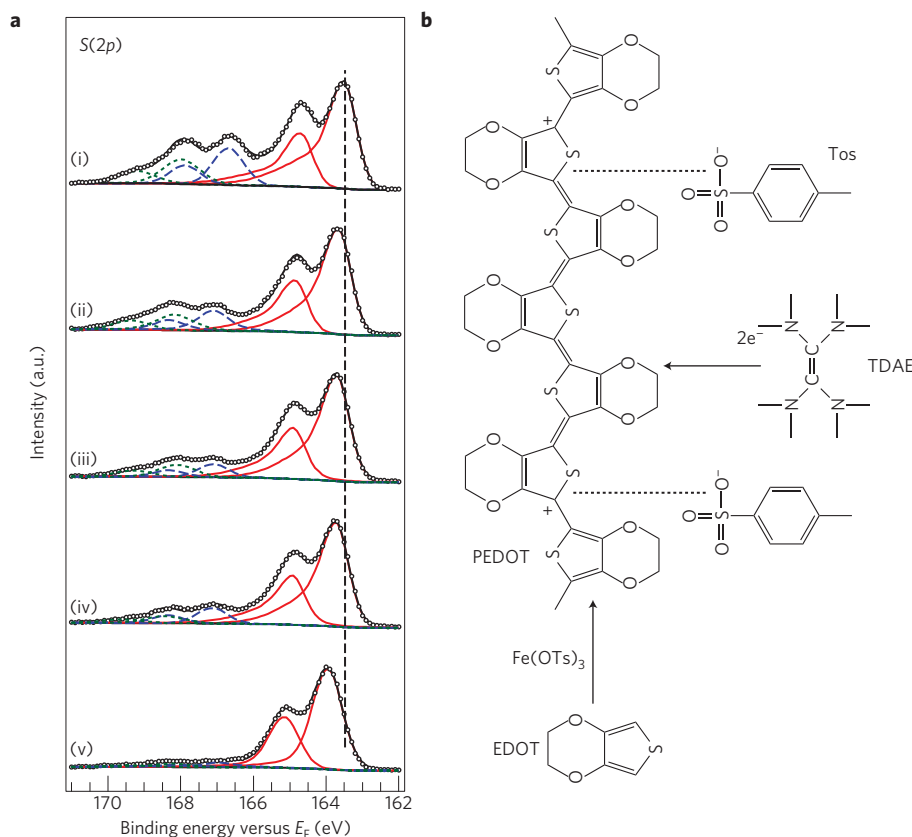


Figure 1 | Chemical structure and photoelectron spectroscopy fingerprint. **a**, $S(2p)$ XPS spectra of the chemically polymerized PEDOT-Tos films. Spectrum (i) is for the pristine p-doped polymers. The two peaks at high binding energy (170–166 eV) are from Tos, whereas the doublet (166–162 eV) is from PEDOT. The ratio of those contributions provides the oxidation level of $36.4 \pm 1.0\%$. The other spectra correspond to PEDOT-Tos reduced by TDAE to reach different oxidation levels: (ii) 28.4%, (iii) 16.2%, (iv) 14.3% and (v) 9.0% with an error about $\pm 1.0\%$. **b**, Sketch of the oxidative polymerization of EDOT by iron tosylate that creates the oxidized form of PEDOT. Also, the exposure with TDAE reduces PEDOT-Tos to decrease the oxidation level.

forms a salt together with tosylate, which is easily removed from the film by rinsing with water. X-ray photoelectron spectroscopy is an ideal tool to follow the reduction of PEDOT-Tos and estimate the oxidation level. The $S(2p)$ signals from the sulphonate group of Tos (166–170 eV) and the thiophene units in PEDOT (163–166 eV) are clearly distinguishable because of the different chemical environment of their sulphur atoms (Fig. 1). The $S(2p)$ electrons in the sulphonate have a high binding energy owing to the presence of three electronegative oxygen atoms. For different TDAE exposure times, the conducting polymer reaches distinct oxidation levels, directly quantified by the ratio between the two $S(2p)$ contributions, as each tosylate acts as a counterion to balance the doping charge carried on the PEDOT chains. Figure 1a displays the X-ray photoemission spectroscopy (XPS) $S(2p)$ signals of (i) the pristine PEDOT-Tos, as well as those of PEDOT-Tos treated with various TDAE doses to reach oxidation levels of: (ii) $28.4 \pm 1.0\%$, (iii) $16.2 \pm 1.0\%$, (iv) $14.3 \pm 1.0\%$, and (v) $9.0 \pm 1.0\%$. After the reduction reaction, PEDOT-Tos is stable in air at room temperature for the range of oxidation levels of interest in this study (see Supplementary Information). XPS characterization in combination with optical absorption spectroscopy is used to estimate the oxidation level of all the samples, using a procedure detailed in the Supplementary Information.

The thermoelectric properties, namely the electrical conductivity and Seebeck coefficient, are measured as a function of the oxidation level of the PEDOT. At room temperature, the charge-transport in conducting polymers is governed by phonon-assisted hopping that allows a positive (bi)polaron to move between sites in response to the vibrations of surrounding atoms. When the polymer is

reduced and the carrier concentration in the film drops, the PEDOT-Tos film conductivity experiences a dramatic change, as it diminishes from about 300 S cm^{-1} at 36% oxidation level down to $6 \times 10^{-4} \text{ S cm}^{-1}$ at 15% oxidation level (Fig. 2a). The Seebeck coefficient α increases by a factor of 20 on exposing the polymer to the TDAE vapour. Initially, at an oxidation level of 36%, α is found to be about $40 \mu\text{V K}^{-1}$, finally reaching its maximum of $780 \mu\text{V K}^{-1}$ at the lowest considered oxidation level. Interestingly, the power factor σS^2 increases by a factor of 6, from $38 \mu\text{W m}^{-1} \text{ K}^{-2}$ for the pristine doped polymer to $324 \mu\text{W m}^{-1} \text{ K}^{-2}$ at the oxidation level of 22%. At this optimum oxidation level PEDOT-Tos is stable in air, while dioxygen is known to spontaneously oxidize the reduced PEDOT (ref. 16).

To verify that the actual power generation can be optimized with the chemical treatment of PEDOT-Tos with TDAE, a single thermocouple was patterned by ink-jet printing. Printing several layers of PEDOT-Tos allows the fabrication of a $3 \mu\text{m}$ -long pillar on a gold electrode, which constitutes the first leg of the thermocouple. The second leg is obtained from a commercial carbon black ink printed on a second gold electrode. Both legs are electrically connected by a top Au electrode to complete the thermocouple. On applying a small temperature gradient between the top and bottom gold electrodes ($\Delta T = 1.5^\circ\text{C}$), the power output of the fabricated devices is measured with different load resistances for various doses of TDAE (oxidation levels), see Fig. 2c. For a given load resistance, the power output reaches a maximum on TDAE exposure and then drops. The maximum power output is 1.13 nW after 10 min TDAE exposure for a load resistance equal to 39Ω (corresponding to the internal resistance of the thermocouple). This evolution of power

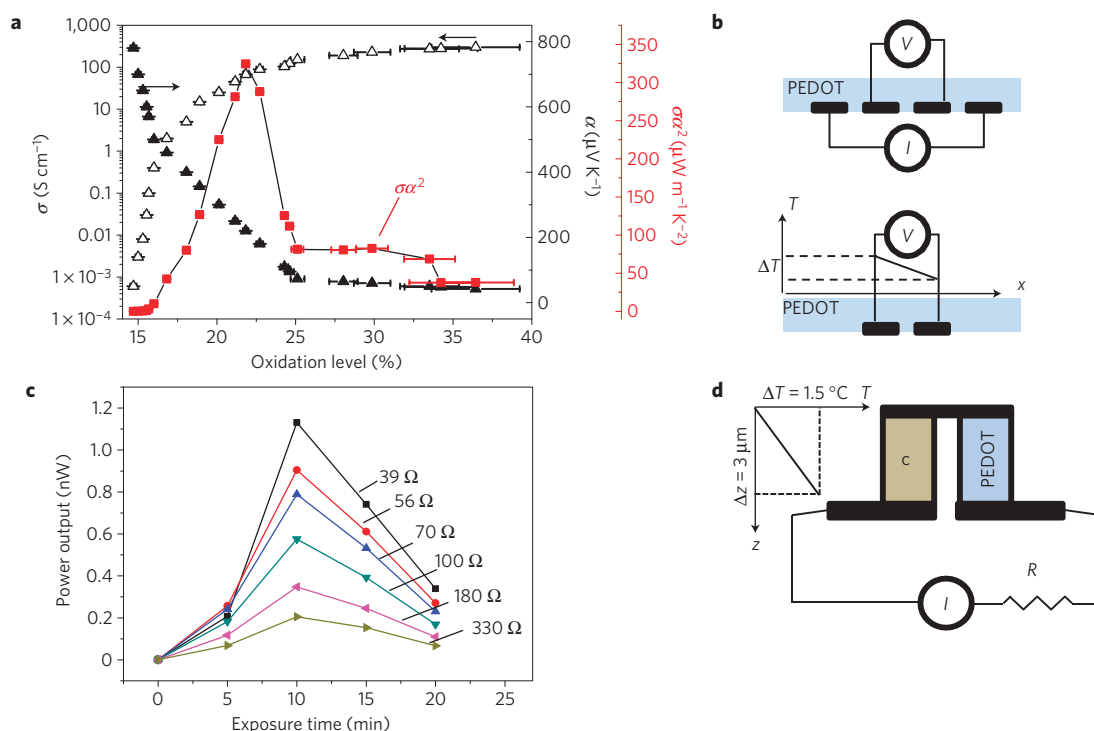


Figure 2 | Thermoelectric properties versus oxidation level. **a**, Seebeck coefficient α (filled triangles), electrical conductivity σ (open triangles) and corresponding power factor $\sigma\alpha^2$ (red squares) versus oxidation level (horizontal error bars are estimated in the Supplementary Information). **b**, Schematics of the four-point probe measurement of the conductivity and the Seebeck coefficient measurement $\alpha = \Delta T/V$ (below). **c**, The measured useful power output of a PEDOT-Tos/carbon thermocouple with different loads versus exposure time to the TDAE vapour. **d**, Schematic of the power output measurement set-up. The thickness of the PEDOT-Tos and carbon legs is about $3 \mu\text{m}$ and the temperature gradient $\Delta T = 1.5^\circ\text{C}$.

versus TDAE exposure time is direct evidence that TDAE modifies the power factor of PEDOT-Tos.

To estimate the order of magnitude of ZT , the thermal conductivity of PEDOT-Tos is measured using the 3ω -technique, now considered as a standard method for thin film characterization^{7,17}. An a.c. current at frequency ω flowing through the thin heater causes Joule heating. The temperature in the heater oscillates with the applied power at twice the frequency of excitation. This oscillation of temperature ΔT implies a variation of the resistance of the heater. Hence the Au heater can also be used as a thermometer by measuring its resistance (depending on the temperature). Thus, ΔT is measured by the third harmonic voltage $V_{3\omega}$ across the heater. This method is particularly appropriate for amorphous materials. Indeed, thin polymer films ($> 10 \text{ nm}$) typically have a thermal conductivity close to the bulk value, as phonon scattering at the film-substrate and film-atmosphere interfaces is limited because of the short intrinsic phonon mean free paths¹⁸. The vertical thermal conductivity $\lambda_{\perp}$ (perpendicular to the plane of the substrate) of a PEDOT-Tos film is $0.33 \pm 0.1 \text{ W m}^{-1} \text{ K}^{-1}$ (see Supplementary Information). The lateral thermal conductivity $\lambda_{\parallel}$ (parallel to the substrate) is also measured using the 3ω -method¹⁹ by comparing the heat transport from wide ($34 \mu\text{m}$) and narrow ($2.3 \mu\text{m}$) insulated metal heaters deposited on the PEDOT-Tos film (Fig. 3a). Figure 3b displays the reduced temperature oscillation ΔT_{1D} of the wide $34 \mu\text{m}$ gold heater, which has a predominant unidirectional (1D) heat propagation perpendicular to the substrate; thus serving as a reference for the vertical thermal conductivity $\lambda_{\perp}$. In contrast, heat propagation parallel to the substrate becomes a significant effect (lateral heat spreading) for the narrow $2.3 \mu\text{m}$ Au heater. The ratio between reduced temperature oscillations of the narrow heater, ΔT_m and ΔT_{1D} , together with the ratio between the width (w) of the heater and the thickness of the film (d) are used to extract the

ratio $\lambda_{\parallel}/\lambda_{\perp} = 1.11$ (see inset, Fig. 3b). A ratio close to one indicates that the heat propagation is almost isotropic, which is attributed to the preparation method of the film, which consists in spin-coating the monomer solution prior polymerization. The lateral thermal conductivity of PEDOT-Tos, $\lambda_{\parallel} = 0.37 \pm 0.07 \text{ W m}^{-1} \text{ K}^{-1}$, together with the measured power factor (Fig. 2a), defines the material figure of merit ZT at room temperature. For conducting polymers, no dependence has been observed between the thermal conductivity and the oxidation state⁴. ZT is reported versus oxidation level in Fig. 3c, with a maximum value of $ZT = 0.25$, the highest ever shown for conducting polymers.

Solution processability of conducting polymers offers a unique advantage for patterning the semiconducting legs of thermogenerators. To illustrate this, a thermoelectric module based on PEDOT-Tos was fabricated. The all-organic TEG consists of 54 legs with lengths of approximately $40 \mu\text{m}$ (Fig. 4a). Epoxy-based polymer cavities were created by photolithography above the bottom gold electrodes. Every other cavity was filled with a solution of the p-type semiconductor, namely an EDOT/Tos/pyridine solution, and then annealed at 110°C to induce the polymerization into PEDOT-Tos. All the remaining openings were filled with the organic conducting salt tetrathiafulvalene-tetracyanoquinodimethane (TTF-TCNQ), which possesses a negative Seebeck coefficient of $-28 \mu\text{V K}^{-1}$ (ref. 20). For processability reasons, the TTF-TCNQ crystalline powder was blended with polyvinylchloride (PVC) in a ratio of 3:1 in toluene. The TTF-TCNQ/PVC blend forms a black paste with a sheet resistivity of about $11 \text{ k}\Omega \text{ cm}^{-1}$ and a Seebeck coefficient of $-48 \mu\text{V K}^{-1}$. The measured maximum power output ($P_{\text{max}} = U^2/(4R_i)$) of the TEG is defined using the inner resistance R_i of the thermogenerator at the corresponding temperature difference and the measured Seebeck voltage U . The results are shown in Fig. 4b. Even though the n-leg and the device geometry are not optimized, this all-organic TEG provides a maximum power output

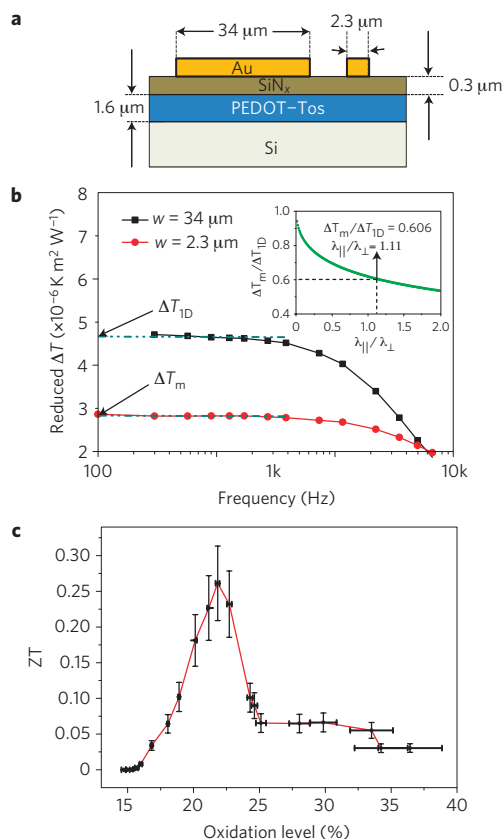


Figure 3 | Thermal conductivity and thermoelectric figure of merit.

a, Sketch of the device to estimate the ratio between the lateral and vertical thermal conductivities for a PEDOT-Tos film. The two gold heaters (thermometers) are both 0.5 mm long and insulated by a thin SiN_x layer deposited by low-temperature chemical vapour deposition ($T < 90^\circ\text{C}$). **b**, The reduced temperature oscillation ΔT_{D} on the 34 μm wide Au line allows the vertical thermal conductivity $\lambda_{\perp}$ to be determined. A reduced temperature is used to account for dimensions and input electrical power, thus making a comparison easier with other studies. The 2.3 μm Au heater is characterized by the reduced temperature oscillations ΔT_{m} . The ratio $\Delta T_{\text{m}}/\Delta T_{\text{D}}$, together with the ratio between the width of the heater, w , and the thickness of the film, d , are used to extract the ratio $\lambda_{\parallel}/\lambda_{\perp} = 1.11$ (see inset). **c**, Values of ZT at room temperature for various oxidation level assuming that the thermal conductivity of PEDOT-Tos is constant and equals $0.37\text{ W m}^{-1}\text{ K}^{-1}$. The vertical error bars mostly originate from the uncertainty in the lateral thermal conductivity (error about 20%).

of $0.128\text{ }\mu\text{W}$ at $\Delta T = 10^\circ\text{C}$. Assuming an optimum packing density of the legs (dashed line in Fig. 4), the expected electrical power generation from such modules can be extrapolated to $0.27\text{ }\mu\text{W cm}^{-2}$ at $\Delta T = 30^\circ\text{C}$ (a temperature gradient possible to achieve in micro-thermogenators²¹). The first target applications considered for this generation of organic thermogenators are energy scavengers for charging supercapacitors to produce sufficient electrical power for low-power wireless and medical sensors introduced on packages or textiles: ‘internet-of-things’²². The low-cost, large-area compatibility and the use of easily recyclable and environmentally friendly materials are crucial to achieve such mass-produced products.

Methods

Synthesis. The chemical polymerization of EDOT in the presence of iron tosylate followed a method described elsewhere¹⁰. 50 μl of 3,4-ethylenedioxythiophene (Sigma-Aldrich) was added to 1 ml of oxidative solution Baytron C, which contained 20% diluted iron (III) *tris*-p-toluenesulphonate in butanol (Baytron C). Pyridine was added as an inhibitor to the solution (0.5 mol per 1 mol of iron tosylate) to control the kinetics of the polymerization. The resulting solution was

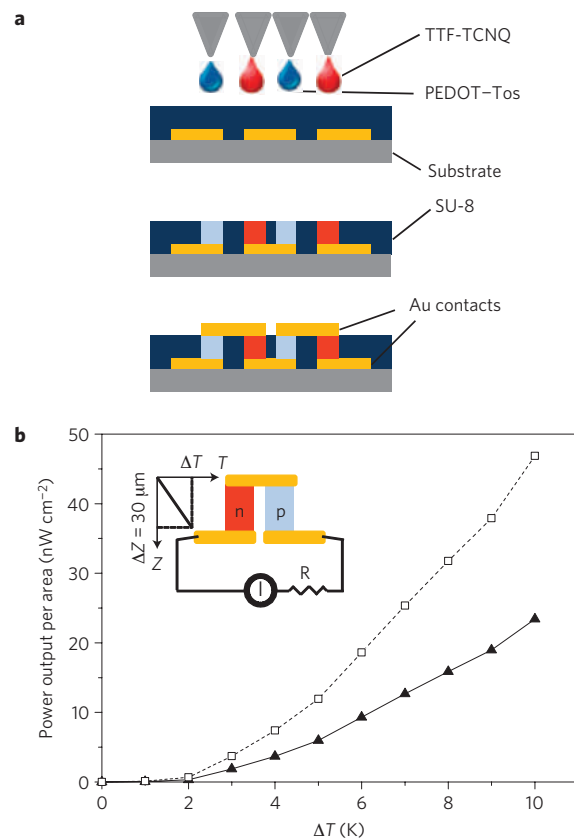


Figure 4 | Manufacturing and power generation of the organic thermoelectric module.

a, Thermogenerator fabrication process steps: Cavities are formed above the bottom Au electrodes using the photoresist SU-8. The cavities are filled with a solution of EDOT and TTF-TCNQ respectively to form the p-legs and n-legs. The TEG is completed by depositing the top Au contacts. **b**, Maximum power output per area of the PEDOT-Tos/TTF-TCNQ TEGs consisting of 54 thermocouples with the following leg dimensions: $25\text{ mm} \times 25\text{ mm} \times 30\text{ }\mu\text{m}$. The packing density for the matrix, defined as the area occupied by the legs divided by the total area of the TEG, is 0.47. The extrapolated limit power output for a packing density of 0.94 is plotted as the dashed line. The inset figure represents a thermocouple formed by one PEDOT-Tos and one TTF-TCNQ leg.

spin-coated on glass substrates at 1,000 r.p.m. for 30 s to obtain a film thickness of about 150–200 nm, as measured with a DEKTA profilometer. The samples were dried and baked on a hot-plate at 110°C for 5 min to finish the polymerization. The PEDOT-Tos samples were washed several times with deionized water and subsequently dried.

The reduction of PEDOT-Tos with tetrakis(dimethylamino) ethylene—TDAE—(Sigma Aldrich, 90% purity) was realized in a nitrogen atmosphere at room temperature. A drop of TDAE was deposited in the vicinity of the PEDOT-Tos film in a closed vessel to ensure a pressure of TDAE close to its vapour pressure. Varying the TDAE exposure time led to various oxidation levels for the conducting polymer.

Characterization. Photoemission experiments were carried out using a Scienta ESCA 200 spectrometer in ultrahigh vacuum with a base pressure of 1×10^{-10} mbar. The measurement chamber was equipped with a monochromatic Al K α X-ray source providing photons with an energy of 1,486.6 eV. The XPS experimental conditions were set so that the full-width at half-maximum of the clean Au 4f_{7/2} line was 0.65 eV. Deconvolution of the S(2p) spectra was carried out on the basis of *a priori* knowledge of the chemical composition, and on the results obtained on thin films of TDAE reported in the literature^{12,15}. During the deconvolution performed on the S(2p) spectra, the energy difference between the S(2p_{1/2}) and S(2p_{3/2}) spin-orbit components was kept at 1.2 eV, and the intensity ratio fixed at 1:2.

The thermopower measurements were performed using the 2D geometry structure reported elsewhere²². The absolute value of the Seebeck coefficient of PEDOT-Tos was calculated for each oxidation level by subtracting the Seebeck

coefficient of the gold electrodes ($S = +1.94 \mu\text{V K}^{-1}$ at 25°C) from the measured relative Seebeck coefficient. The temperature difference was monitored using an infrared camera FLIR A325 (thermal sensitivity $<50 \text{ mK}$). The accuracy of the measurements was verified by k-type thermocouples. The electrical conductivity measurements were done by the standard 4-point probe method (four patterned Au lines on glass) using a semiconductor characterization system (Keithley 4200-SCS).

The thermal conductivity of PEDOT–Tos was measured using the differential 3ω -method. Using an arbitrary waveform generator (33250A), a sinusoidal input current (at different frequencies) was applied to a bridge circuit having the gold heater in one of its branches. The 3ω -voltage was measured by a lock-in-amplifier (SR850 from Stanford Research) at its two differential inputs, while the bridge was balanced by a variable resistor to minimize the first harmonic voltage. All the measurements were carried out inside a cryogenic probe station (Janis Research) under vacuum. The thermometer was calibrated from the evolution of the resistance of the heater versus temperature in a climate chamber.

Device fabrication. For the single thermocouple fabrication, two gold bottom electrodes, each $2 \times 5 \text{ mm}$ and 2 mm apart, were evaporated onto a silicon substrate. The thermocouple legs were deposited using an inkjet printer (Dimatix Material Printer 2800). As the PEDOT–Tos is not soluble, we used an original approach: a solution of the EDOT monomer including the oxidant ($\text{Fe}(\text{Tos})_3$) was printed on the warm Au electrode. The spontaneous polymerization at room temperature in the cartridge is slowed down significantly by adding pyridine. As soon as the drops of solution land on the warm electrode (35°C), the polymerization starts. The bottom electrode was insulated using a $3 \mu\text{m}$ -thick polystyrene layer before evaporation of the top electrode. To achieve a reasonable thickness for the thermogenerator legs, each pattern was repeated 30 times, resulting in a $3 \mu\text{m}$ -thick PEDOT–Tos leg.

For the thermoelectric module, 55 equally spaced 50 nm -thick ($3 \times 1.5 \text{ mm}$) Au electrodes were deposited on top of the Si support wafer. The photoresist SU-8 50 was used to create $45\text{--}50 \mu\text{m}$ -deep cavities and baked for 10 min at 110°C . The cavities (each $1 \times 1.5 \text{ mm}$) were patterned by photolithography and served as containers for the organic semiconductors. The p-type legs were manually filled with EDOT/Tos solution (with weight ratio: 0.24 g (EDOT), 0.4 g (Tos), 0.04 g (Pyridine) in 1.2 ml butanol) and annealed simultaneously at 110°C . The process was repeated several times to achieve thicker PEDOT–Tos legs; each layer was rinsed with deionized water. The other legs were filled with the TTF-TCNQ/PVC blend, characterized by a salt to binder weight ratio 3:1. To finalize the device, the top Au interconnects were deposited in vacuum. The temperature difference between the top and bottom electrodes of the thermogenerator was measured with the calibrated infrared camera.

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Author contributions

X.C. and A.M. initiated the study. O.B. performed the conductivity and thermopower measurements and fabricated the thermogenerators. Z.U.K. developed the 3ω -technique to measure the thermal conductivity. S.B. and M.F. performed the photoelectron spectroscopy measurements. O.B., Z.U.K. and X.C. prepared the manuscript. X.C. and M.B. coordinated the project. All authors revised and approved the manuscript.

Additional information

The authors declare no competing financial interests. Supplementary information accompanies this paper on www.nature.com/naturematerials. Reprints and permissions information is available online at <http://www.nature.com/reprints>. Correspondence and requests for materials should be addressed to X.C.